Fine Tuning

- Full Fine Tuning (No Compute help) → Take a ready model & completely retrain it.
- Semi Fine Tuning :- ~~Full~~ Freeze most train few.
- Direct :- Directly Use readymade model
  - No Compute → easiest.

Ease of Compute :- Full < Semi < Direct

Accuracy (General) :- Full > Semi > Direct

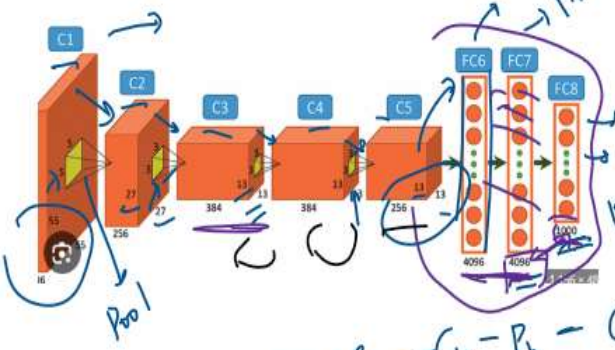
① Images

② Language (LLM's)

→ Depends on situations

Alexnet

5 Convolutions



Input

HL (one hidden layer)

O/P

O/P

1000 Class classifiers

Pool

$C_1 - P_1 - C_2 - R_2 - C_3 - R_3 - C_4 - P_4 - C_5 \rightarrow$ Flatten

W → connection

B → Neuron

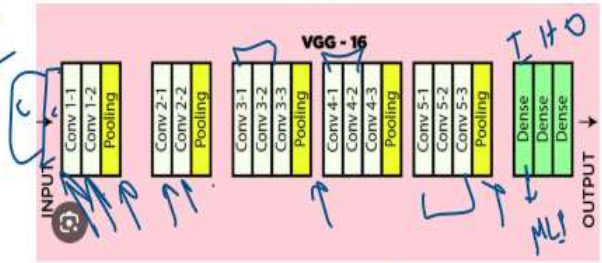
$W = 4096 \times 4096 + 4096 \times 1000$

$B = 4096 + 1000$ (No bias at input)

C-P

C-C-P

VGG-18



FC

ML!

VGG > Alex

"Vision Transformer"

Very heavy

Alex

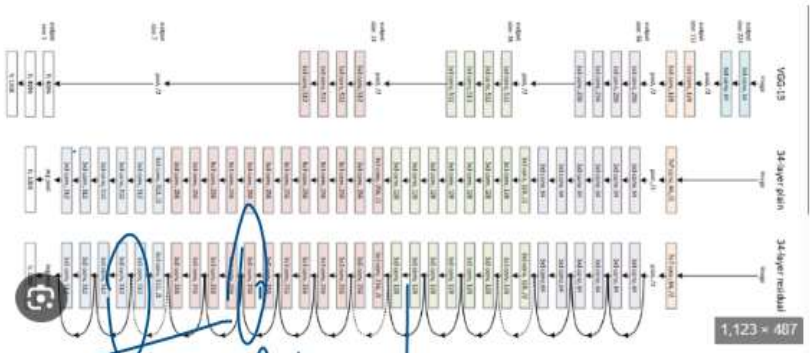
Resnet 50

270

C-P-C1-:

VGG

Resnet



C1-P1-C1-P.

Residual Connection

$x = 5$

$f(x) = x^2$

$f(5) = 25$

5 → f(x) → 25

x → Wx+b → y

5 → 2×5+1 = 11 → 2y+5 → 27

→ filter — FM — P/C — FM

" More deeper I go I lose the input more

with Residual Connection . Transform while retaining

ial data

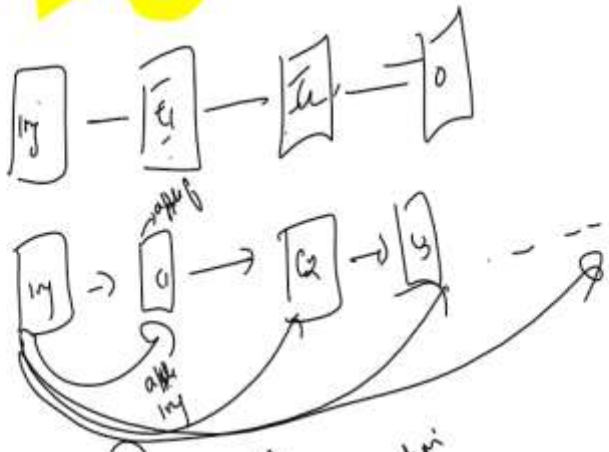
$y = f(x) + x$

$x = 5 \quad f(x) = x^2$

$\Rightarrow y = 25 + 5$

the ...

" Residual connection ... train deep networks "



$y = f(x) + x$ o/p i/p Image with connection

$y = f(x) - x$

$y = f(x)$ without RC

$w_t = w_t + \ldots$

$w_t = w_{t-1}$

$\sigma = 0 - 0.25$

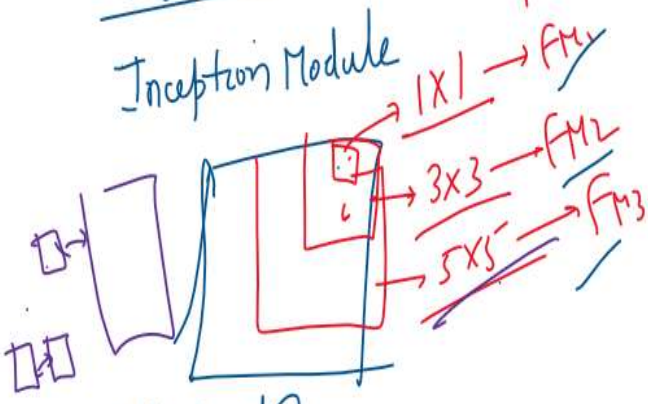
$\ldots 0.03 \times 0.21$

" Whisper "

" Wave to Vec " (CNN + Trans)

GoogleNet

Inception Module



"Parallely put multiple filters of different size to learn multiple dynamics of the image"

what to do?

fm1 fm2 fm3

"In each feature map ensure H & W is same"

$$O = \frac{I - K + 2P}{S} + 1$$

(H & W)

1	2
3	4

1	1
1	1

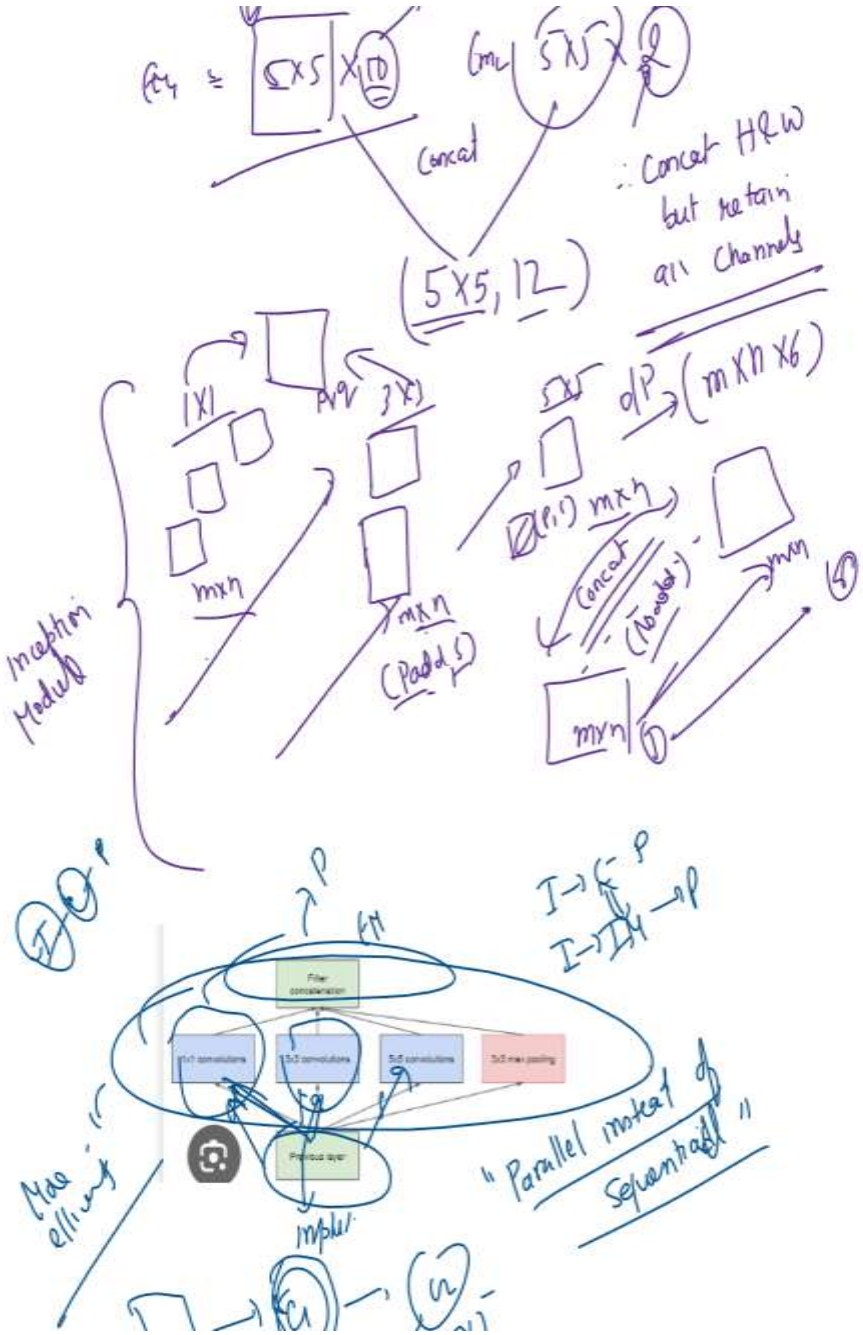
$= \begin{bmatrix} 2, & 3 \\ 4, & 5 \end{bmatrix}$

RGB → (H, W, Channels)

Kernels

No Need of Same be-

1X1 in 3X3


5X5
Concat
(5X5, 12)
∴ Concat H&W but retain all channels
1X1
3X3
5X5
m×n
(Padd 5)
(m×n×6)
(concat)
(Naaded)
inception module

Filter concatenation
1x1 convolutions
3x3 convolutions
5x5 convolutions
3x3 max pooling
Previous layer
"Parallel instead of Sequential"